

TOPIC:-

Activity 1:

Make a presentation of biogas plant and the process of gas production.

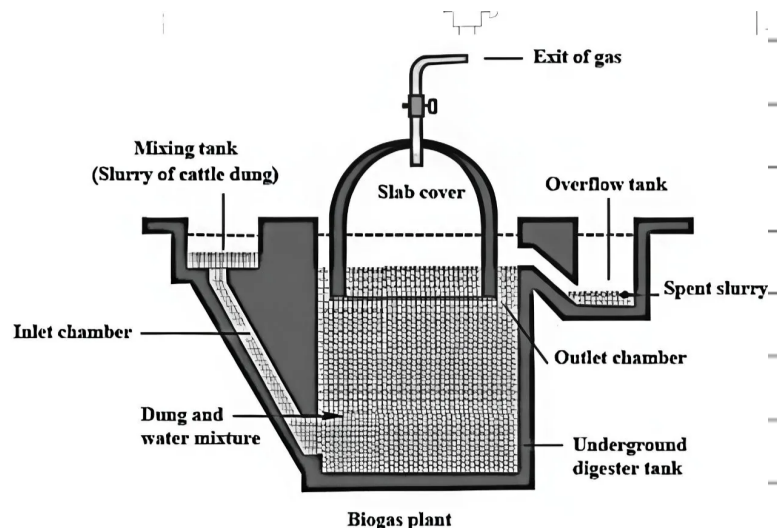
INTRODUCTION:-

This project is based on biogas which is increasingly used as an alternative source of energy. Human civilization has been using energy in different forms from the very first day of their existence.

At current rates of consumption, known reserves of petroleum will be gone in about thirty five years, natural gas in fifty two years, and coal in some two hundred years. Hence, there is a need to look for alternative sources of energy including in rural areas of India where majority of people live.

Biogas plant is one of the alternatives which are efficient, cost effective and hygienic. This project tells about the construction, working, cost, advantages and demand of biogas in coming decades.

► WHAT IS BIOGAS:-



Biogas is a flammable gas that occurs from the fermentation of biomass in biogas plants.

Biogas typically refers to a gas produced by the biological breakdown of organic matter in the absence of oxygen.

Biogas originates from organic material and is a type of biofuel.

Biogas is a renewable source of energy.

Organic waste such as dead plant and animal material, animal dung, and kitchen waste can be converted into a gaseous fuel called biogas.

► HOW IS BIOGAS GENERATED:-

Biogas is practically produced as landfill gas or digester gas.

Biogas can be produced using anaerobic digesters.

These plants can be fed with energy crops such as maize silage or biodegradable wastes including sewage sludge and food waste.

► PROCESS WORKING:-

During the process, an air tight tank transforms biomass waste into methane producing renewable energy.

Landfill gas is produced by wet organic waste decomposing under anaerobic conditions in a landfill.

The waste is covered and mechanically compressed by the weight of the material deposited from above.

This material prevents oxygen exposure, thus allowing anaerobic microbes to thrive.

The gas builds up during this process.

► WHAT IS BIOGAS PLANT:-

A biogas plant is the name often given to an anaerobic digester that treats farm wastes or energy crops.

Biogas can be produced using anaerobic digesters.

► Components of biogas plant:-

Concrete tank digester. The bio wastes are collected here.

Floating cover. It is placed over the tank and keeps on rising as gas increases.

Outlet. The outlet is connected to a pipe through which gas is collected and further used.

Slurry collector. Cow dung is added which contains bacteria that act on bio waste and produce methane gas.

Sludge collector. The spent slurry is removed and can be used as fertilizer.

► WHAT DOES BIOGAS CONTAIN:-

- An average biogas contains
- 55 to 80 percent methane
- 20 to 40 percent carbon dioxide
- Trace gases including hydrogen sulphide and nitrous oxide

► ADVANTAGES OF BIOGAS PRODUCTION:-

- Agricultural industrial and municipal wastes can be converted into biofuel.
- Leftover residue of biogas formation can be used as biofertilizer.
- Waste materials that would cause environmental pollution are reduced.
- It is a good alternative for electricity and cooking in rural areas

► DISADVANTAGES OF BIOGAS:-

- Initial setup cost of biogas plant is high.
- Requires regular supply of organic waste.
- Gas production decreases in cold weather.
- Requires proper maintenance and handling.

► USES OF BIOGAS

- Biogas is used for cooking in homes.
- It can be used to generate electricity.
- It is used as fuel for vehicles after purification.
- It can be used for heating and lighting purposes.

► ENVIRONMENTAL IMPACT:-

- Reduces air pollution by replacing fossil fuels.
- Helps in managing organic waste efficiently.
- Reduces greenhouse gas emissions like methane release into atmosphere.
- Produces eco friendly fertilizer as byproduct.

► COST AND MAINTENANCE:-

- Small scale biogas plants are affordable for rural areas.
- Government subsidies are available in many regions.
- Requires regular cleaning and feeding of waste.
- Maintenance cost is low after installation.

► FUTURE SCOPE OF BIOGAS:-

- Increasing demand for renewable energy sources.
- Can play a major role in rural development.
- Helps in achieving sustainable development goals.
- Can reduce dependence on fossil fuels.

CONCLUSION:-

Biogas is a clean and renewable source of energy. It helps in reducing pollution and managing waste effectively. With proper use and development, biogas can become an important energy source for the future, especially in rural areas.

REFERENCES:-

- NCERT. Science Textbook for Class 8. Chapter Coal and Petroleum.
- NCERT. Science Textbook for Class 10. Chapter Sources of Energy.
- Ministry of New and Renewable Energy, Government of India.
- National Biogas and Manure Management Programme.
- MNRE Official Website www.mnre.gov.in
- Energy Education Website www.energyeducation.ca
- Encyclopaedia Britannica www.britannica.com